



PSYCH 201B

Statistical Intuitions for Social Scientists

Course Wrap-Up

A few last logistics...

- **Today:** Please make sure to commit and push 04_notebook.py
- **Tues Mar 17 (Midnight):** Final project deadline
 - If you've diverged from your original proposal, but spoke with me about it already, you *do not* have to update your proposal.qmd
 - Make sure to read the final project page on the course website to review the submission checklist
 - Using R is totally fine if you prefer, but make sure to update your README.md with proper instructions
 - If you're unable to share raw data, please try to share derivatives. If you can't share either that's ok! Just let us know in the README.md (e.g. dataset from study-in-progress)
- **Fri Mar 20 (midnight):** deadline to push updates to any previous HWs/labs

Our journey together...

By the end of the term I want you to feel
empowered and **formidable** in your
Python, analytic, and statistical skills!

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empowered and **formidable** in your
Python, analytic, and statistical skills!



~~How we'll learn~~ **How** you've learned

- 13 lectures (slides)
- 5 lab assignments
 - 14 notebooks (including challenges)
- 2 HW assignments
- 1 Final Project

~~Tools we'll use~~ **Tools** you've learned

- Python and the scientific Python ecosystem:
- `polars` - DataFrames & Tidy data-analysis
- `seaborn` - Statistical data visualization
- `bossanova` - Modeling 🖐️
- `scipy` - basic stats, bootstrapping, permutation
- `numpy` - linear algebra

~~Tools we'll use~~ **Tools** you've learned

This took me 3+ years in grad-school!

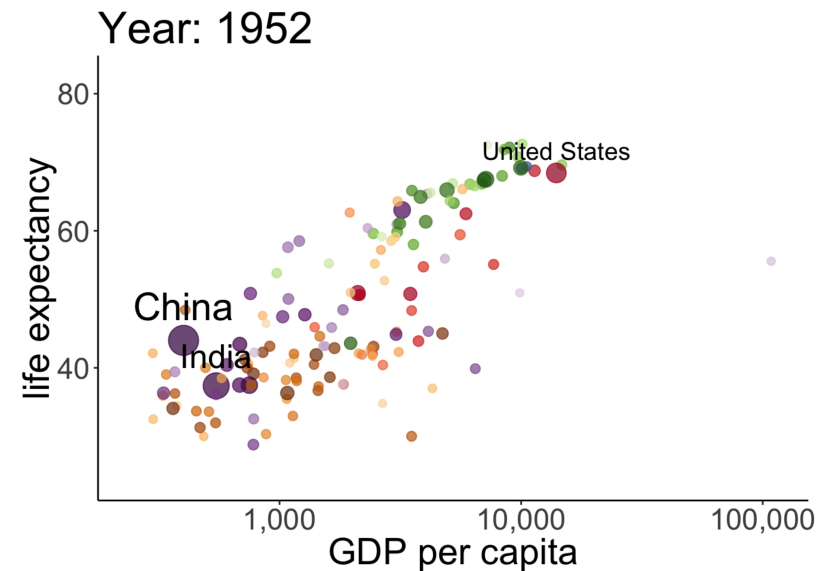
~~Skills you'll learn~~ Skills you've learned



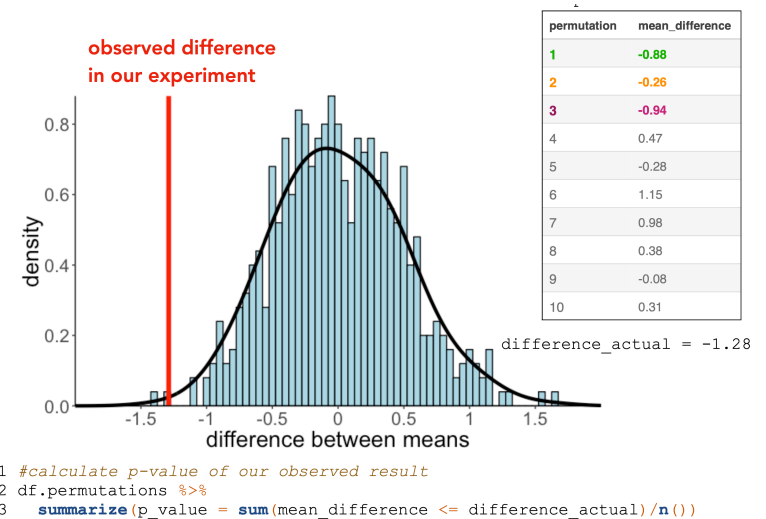
Wrangling Data

1. Define H_0 as Model C (compact) and H_1 as Model A (augmented)
2. Fit model parameters to the data
3. Calculate the proportional reduction of error (PRE) in our sample
4. Decide whether the augmented model is **worth it** by comparing the observed PRE in our sample to the sampling distribution of PRE (assuming that H_0 is true)

Building & Comparing models



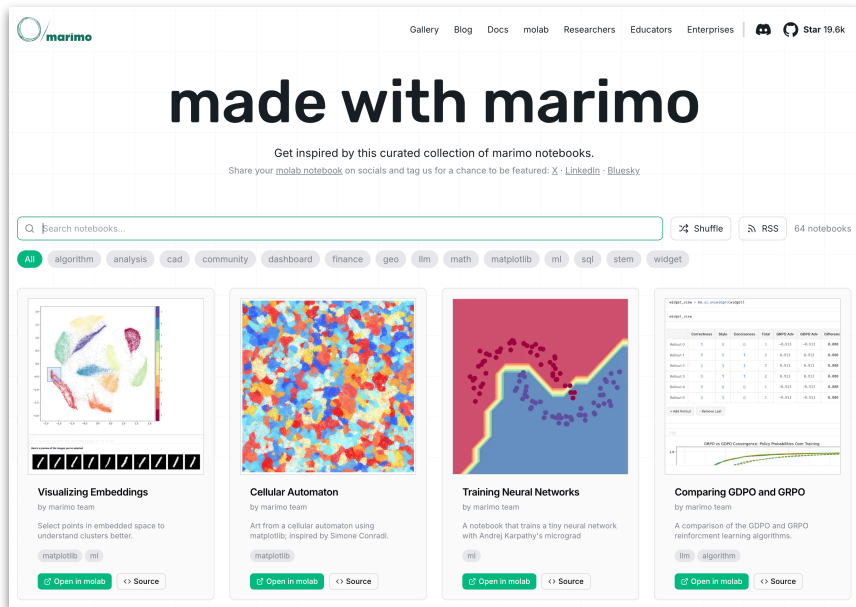
Visualizing Data



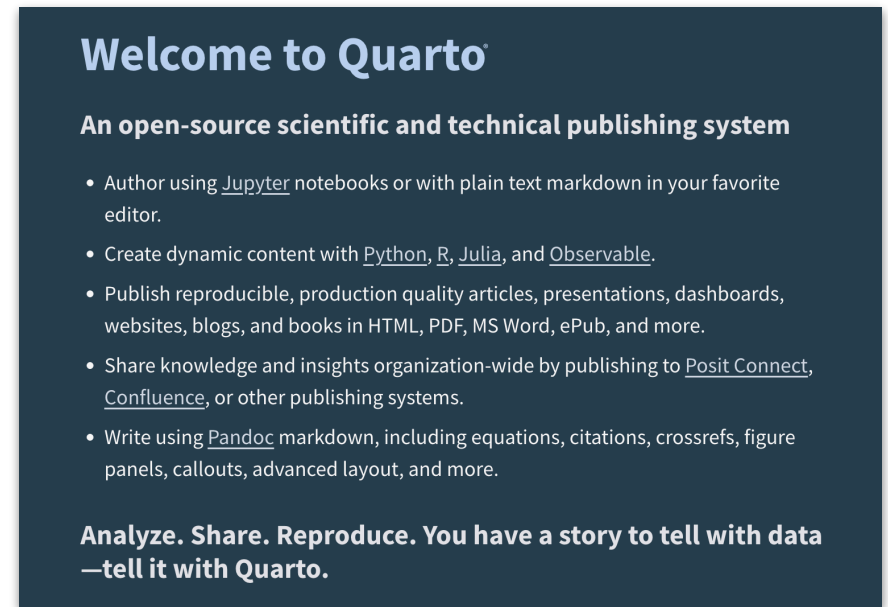
p-value = .002

Performing inference

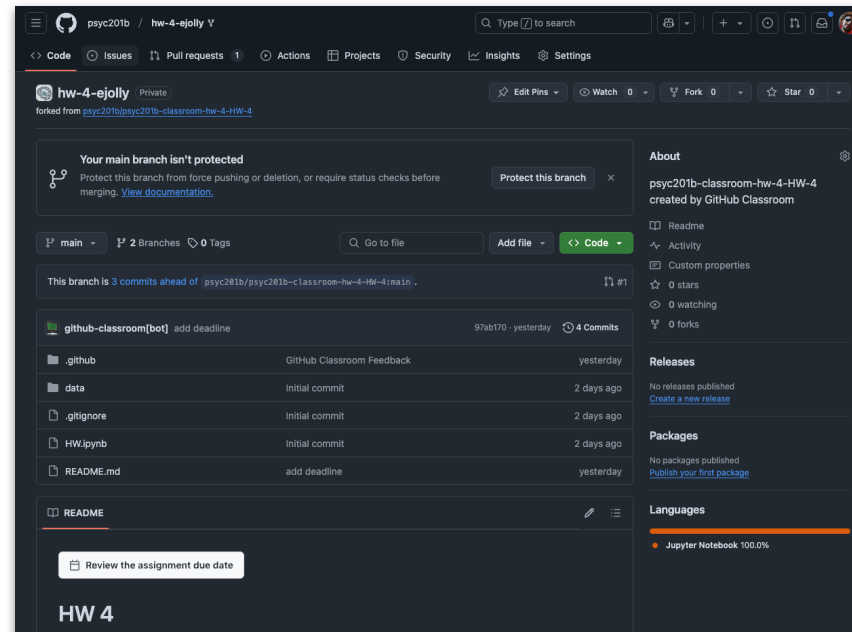
Open and Reproducible science



Marimo Notebooks



Quarto Documents



Git/Github

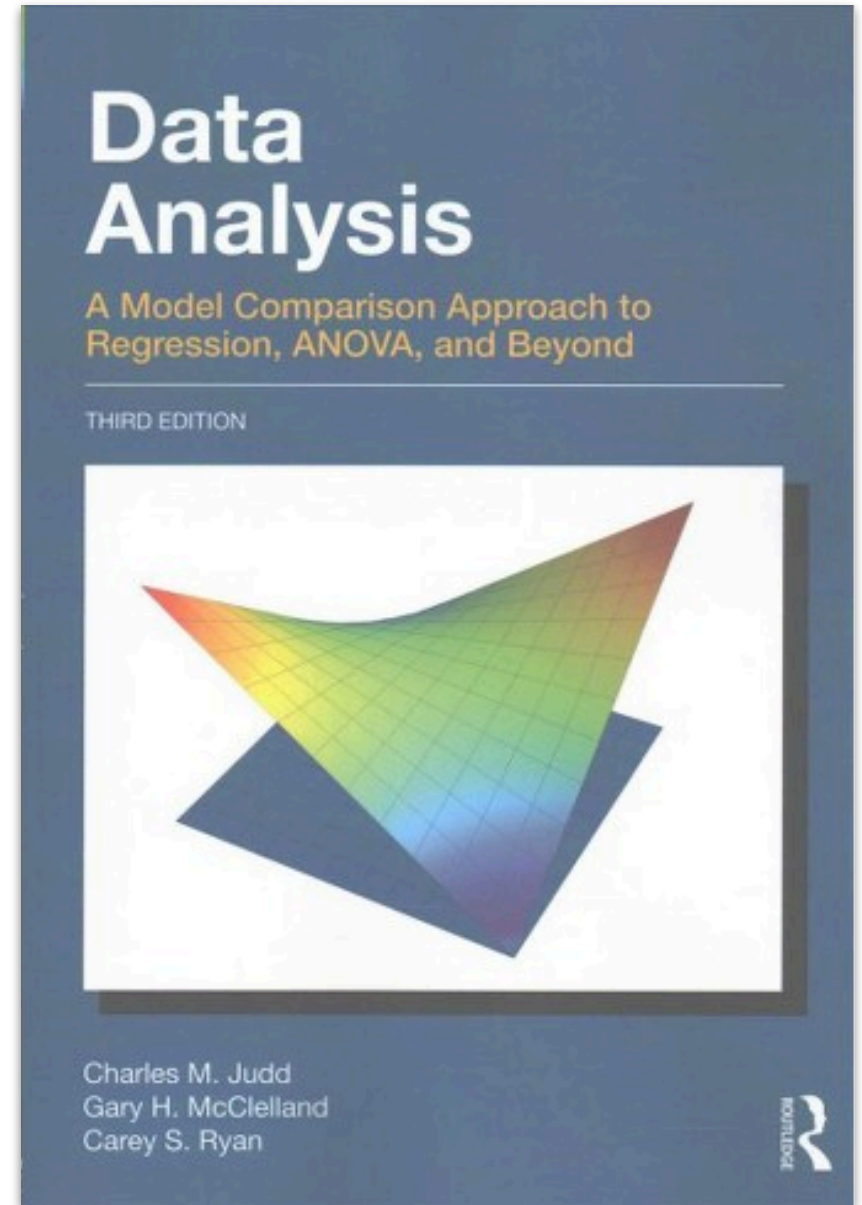
The four fundamental intuitions

- **Aggregation**
 - Gives us a model
- **Sampling**
 - Tells us where it applies
- **Uncertainty**
 - Keeps us honest
- **Learning**
 - Forces iteration

Statistics is **not** about being right

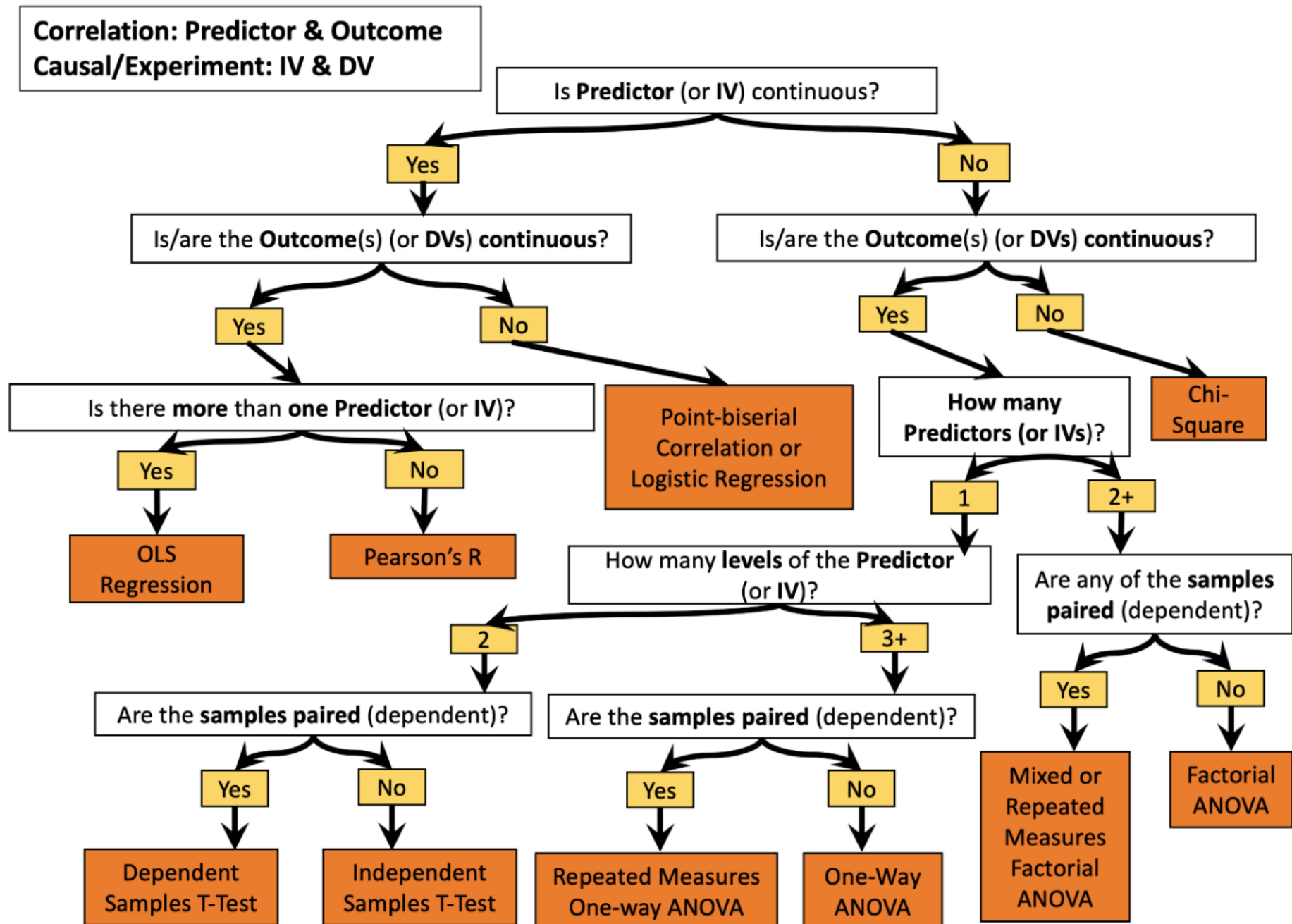
It's the process of becoming **less wrong over time**

Model comparison approach

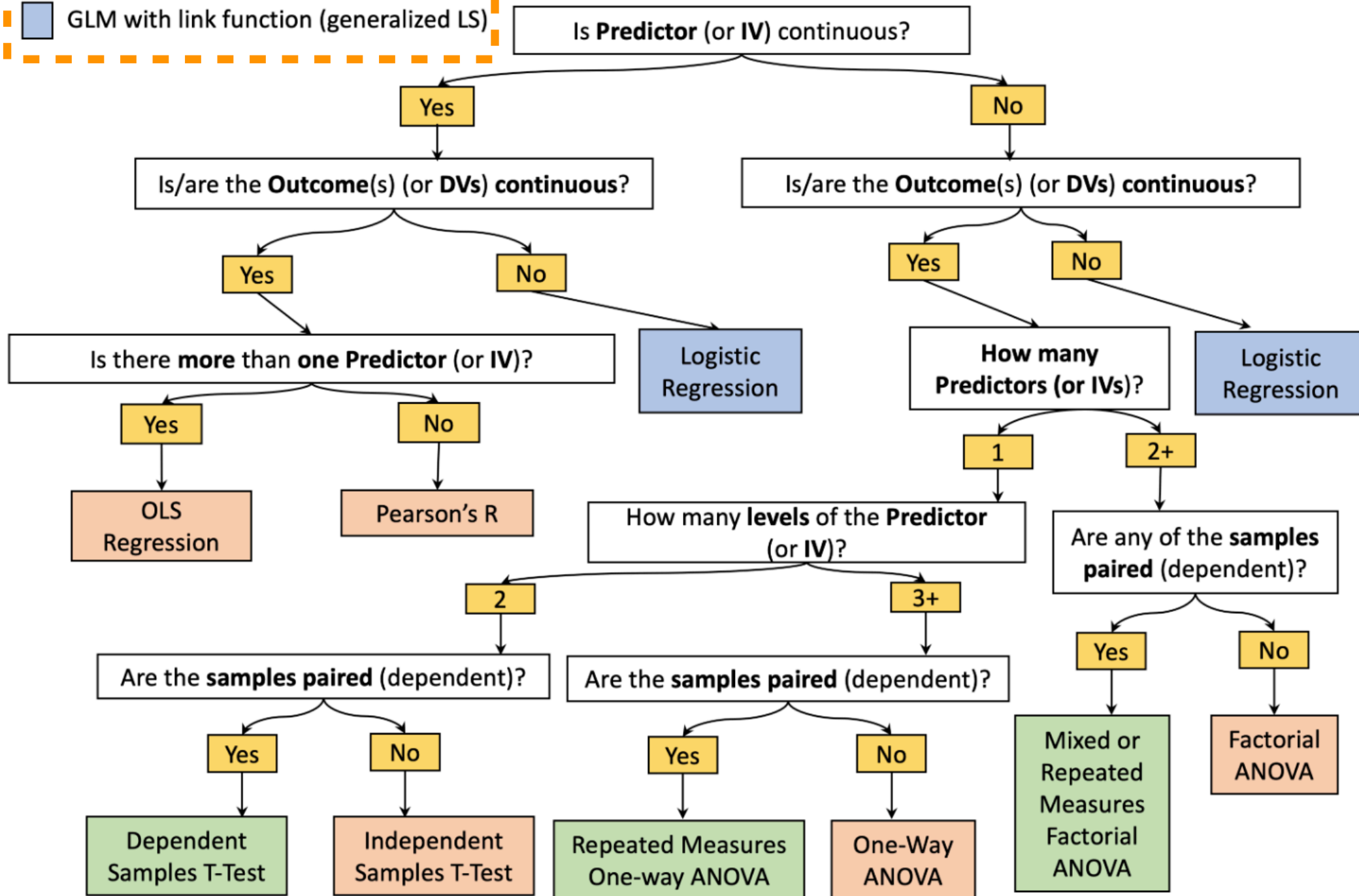
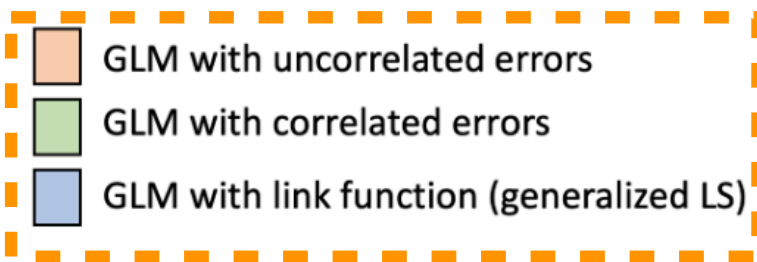


Judd, C. M., McClelland, G. H., & Ryan, C. S. (2011). *Data analysis: A model comparison approach*. Routledge.

Instead of this...



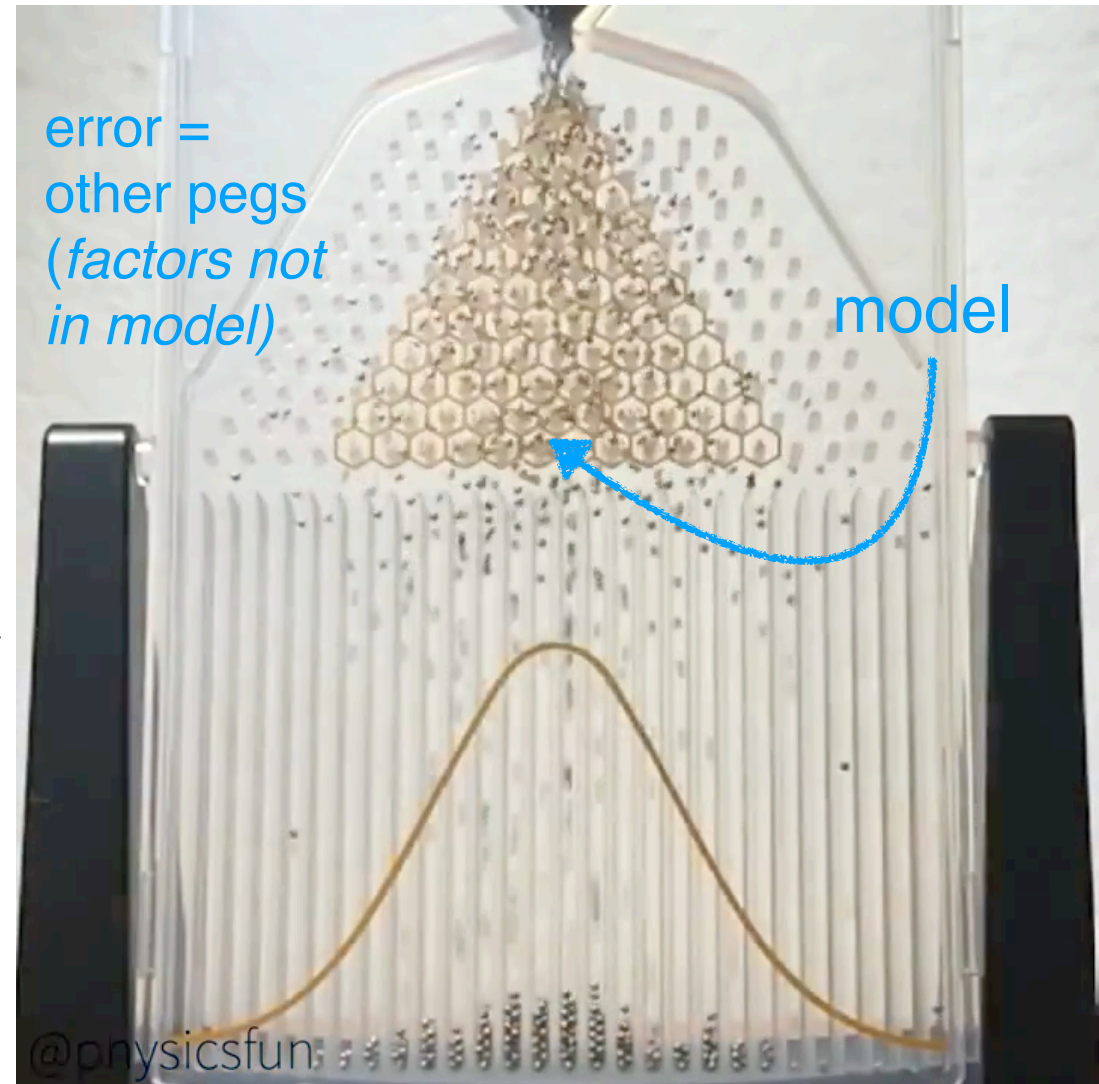
...we think like this



GLM Assumptions

Errors should be independent & identically **distributed (i.i.d)**

- Normality
- Additivity & Linearity
- Homoscedasticity
- No perfect multi-collinearity
- Independence of errors



data = beads

Linear Mixed Models allow us to account for violations of independence

$$y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{ij} + \epsilon_{ij}$$

Fixed effects

Varying Effects

y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

independent variables
explanatory variables
regressors
predictors/features
design matrix
input variables

X

$$\begin{bmatrix} \text{---} \vec{x}_1 \text{---} \\ \text{---} \vec{x}_2 \text{---} \\ \vdots \\ \text{---} \vec{x}_n \text{---} \end{bmatrix}$$

β

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

coefficients
betas
slopes
effects
parameter-
estimates

Linear Mixed Models allow us to account for violations of independence

$$y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{ij} + \epsilon_{ij}$$

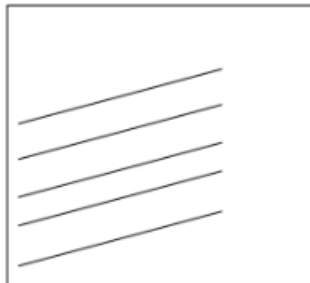
Fixed effects

Varying Effects

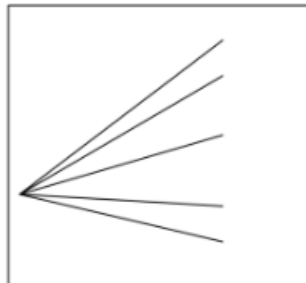
The inference we want to make about the population from our sample

The variation/uncertainty we want to generalize over when making that inference

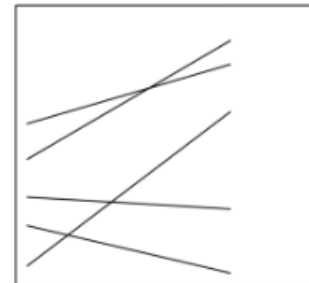
Varying intercepts



Varying slopes

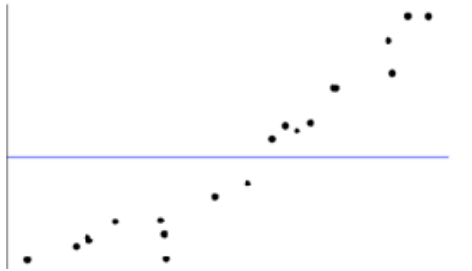


Varying intercepts and slopes



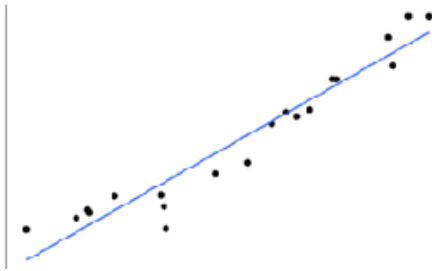
We test hypotheses by comparing models

Which model fits data best?



Compact model

$$\text{model}_C: Y_i = \beta_0 + \text{ERROR}$$



Augmented model

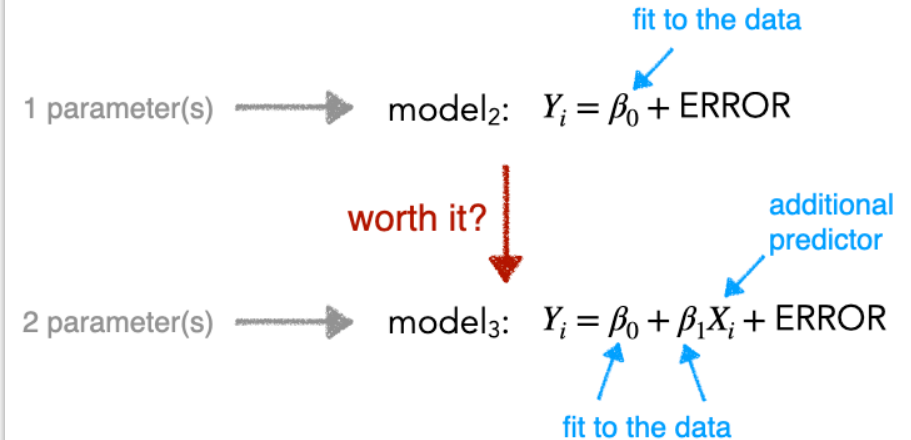
$$\text{model}_A: Y_i = \beta_0 + \beta_1 X_i + \text{ERROR}$$

$$\text{ERROR}(C) \geq \text{ERROR}(A)$$

Proportional reduction in error (PRE)

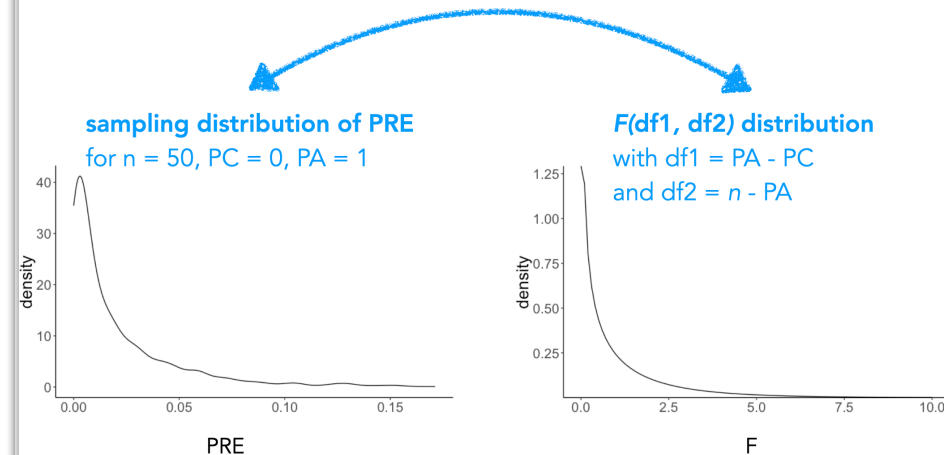
$$\text{PRE} = \frac{\text{ERROR}(C) - \text{ERROR}(A)}{\text{ERROR}(C)}$$

The **worth it?** question



Decide whether it's **worth it**

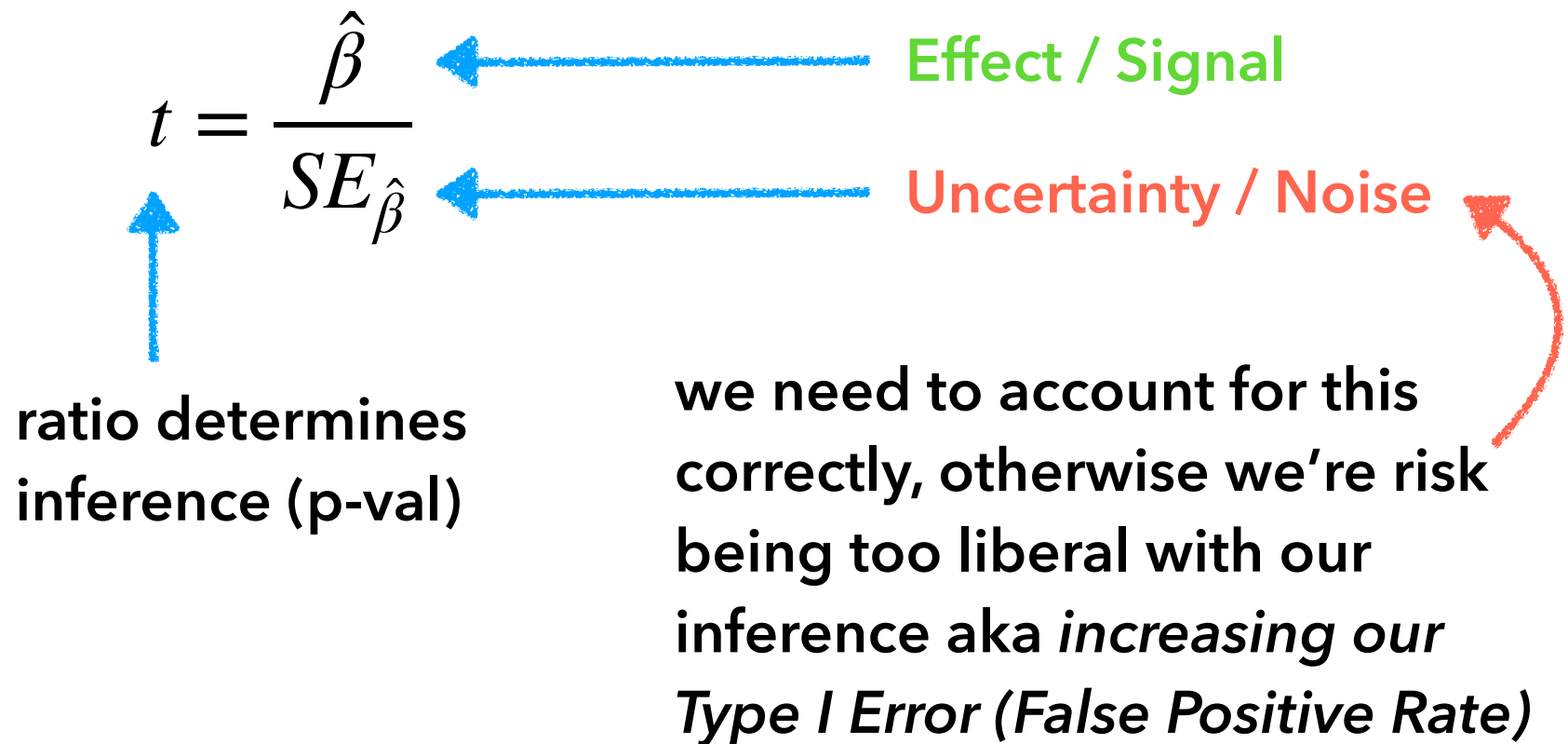
Sampling distribution of PRE = F Statistic



F-statistic: **ratio** of variances/errors

We use **uncertainty** and a **null distribution** to perform **parameter inference**

...which is proportional to nested model comparison



BUT Statistical significance is NOT practical significance

Statistical Power is our **Sensitivity** to true hypotheses

H_0 : Not pregnant H_1 : Pregnant

Ways to
be wrong

false
positive

Type I error

$p(\text{reject } H_0 | H_0 \text{ is true})$

α

false
negative

Type II error

$p(\text{not reject } H_0 | H_1 \text{ is true})$

β

Ways to
be right

true
positive

Sensitivity

$p(\text{reject } H_0 | H_1 \text{ is true})$

Power

$1 - \beta$

true
negative

$p(\text{not reject } H_0 | H_0 \text{ is true})$

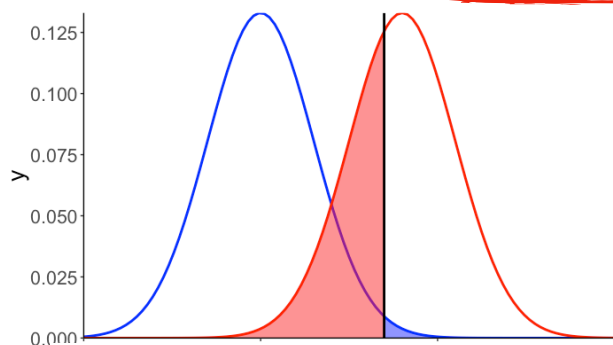
Specificity

$1 - \alpha$

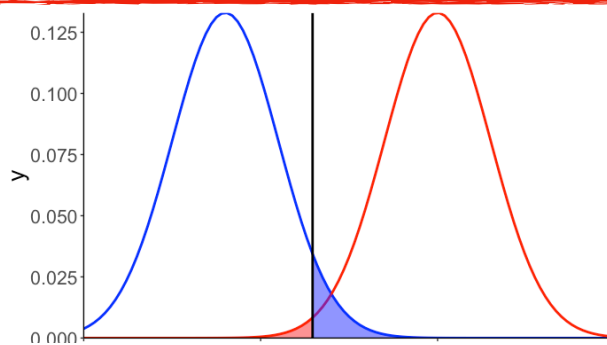
We can use **Simulation** to turn the knobs that affect power

Recipe

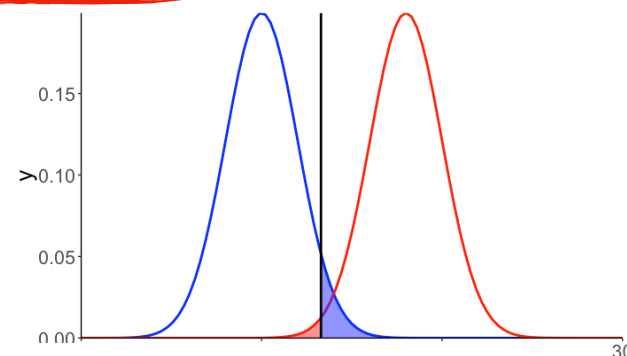
- pick: α , n , *effect-size*
- simulate a large number of data sets of size n with the specified *effect-size*
- for each data set, run a statistical test to calculate the *p-value*
- calculate **sensitivity/power**: proportion of times we *correctly rejected* H_0 (given that H_1 is true)



α threshold



effect size



sample size

We can employ different **resampling approaches** for inference

- **Monte-Carlo (direct) simulation**
 - If the world works like I assume, how likely is my estimator?
- **Bootstrapping**
 - Can the data themselves tell me about the uncertainty of my estimator?
- **Permutation**
 - How likely is this estimator under randomization?
- **Cross-validation**
 - How well does my estimator predict new data?

We use marginal effects to bridge cultures

Explanation

↔

Prediction

Classical inference	Theory			Out-of-sample generalization
	null-hypothesis testing & multiple comparisons	bias-variance decomposition	Vapnik-Chervonenkis dimensions & curse of dimensionality	
	degrees of freedom df		hypothesis space $\mathcal{H}$	
	asymptotic consistency		finite-sample theorems	
	Invalidations of inferential process			
	double dipping/circular analysis	post-selection inference	data snooping/peeking	
	Outcome metrics			
	(in-sample) p values sensitivity/specificity effect size/power	explained variance metrics AUC/ROC curve confidence intervals	out-of-sample prediction accuracy/precision/recall/F1 scores learning curves certainty estimates via bootstrap	
	Representative methods			
	Student's t -test F -test ANOVA Binomial test χ^2 -test linear regression	General(ized) Linear Model GLM	support vector machines LASSO/ridge regression/elastic net logistic regression nearest neighbors random forests kernel methods („deep“) neural networks	

We use a systematic process when statistical modeling

1. Specify your question of interest
2. Identify or collect the appropriate data
3. Prepare (tidy) the data for analysis
4. Determine the appropriate model
5. Fit, inspect, and evaluate the model and its diagnostics
6. Test your hypothesis as a comparison between models
7. Inspect your model parameters & estimation uncertainty
8. Justify your inferences & validate assumptions

Departing “wisdom”...

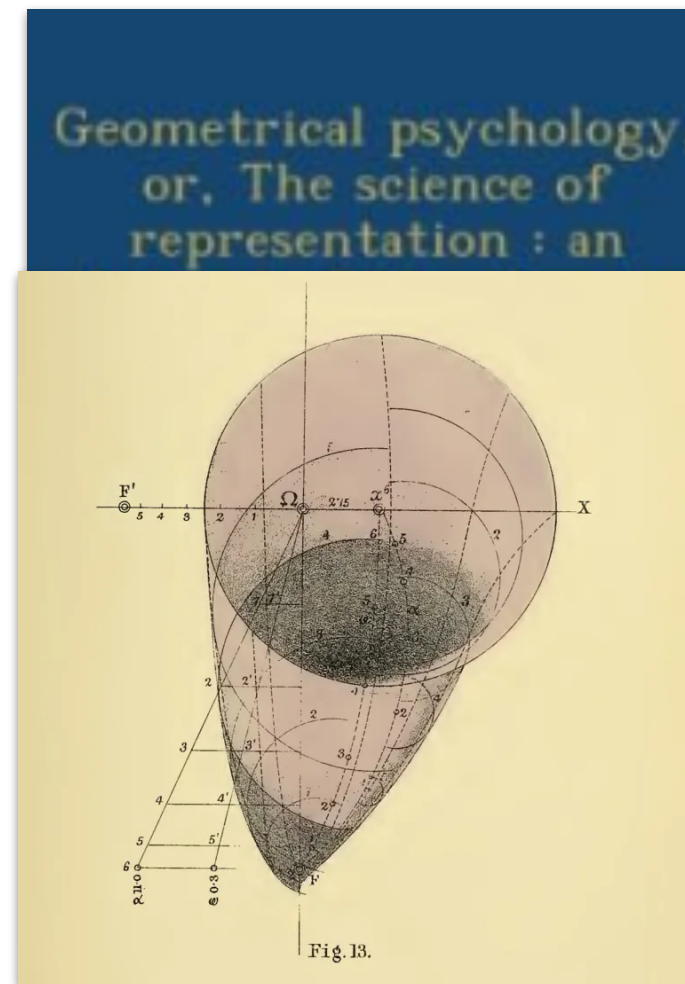
1) Keep improving your statistical thinking

“Statistical thinking will one day be
as necessary for efficient citizenship
as the ability to read and write.”

~ Samuel S. Wilks, 1951
of χ^2 and the log-likelihood ratio!

*“What makes a ~~mathematician~~ **statistician** is NOT technical skill or encyclopedic knowledge...*

*...but an **insatiable curiosity** and desire for simple beauty”*



~ Paul Lockhard 1887
(from Benjamin Betts)

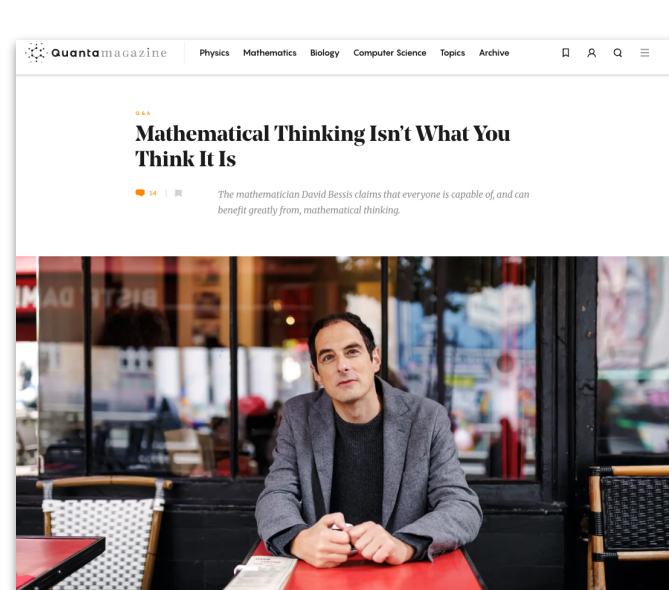
Q & A

Mathematical Thinking Isn't What You Think It Is

🗨️ 14 | 📄

The mathematician David Bessis claims that everyone is capable of, and can benefit greatly from, mathematical thinking.





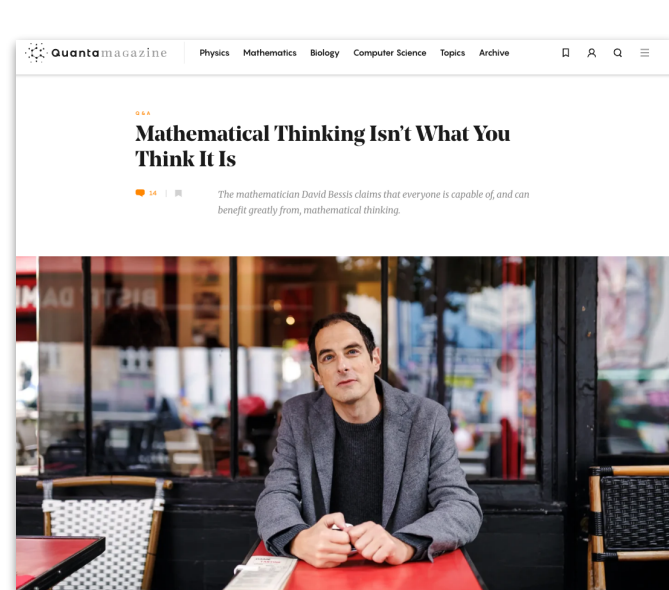
So how can one get better at math?

*“Whenever you **spot a disconnect** between what your gut is telling you and what is supposed to be rational, it’s an important **opportunity to understand something new**. And then you can start this game of back-and-forth:*

***Can you** articulate your gut instinct and **place it within a rational discussion**?
If there’s still a disconnect, **can you visualize why**?*

*As you **play that game**, your imagination will gradually reconfigure. And in the end, if you’re persistent, **your instinct and your reason will align**...this is **mathematical thinking**”*

<https://www.quantamagazine.org/mathematical-thinking-isnt-what-you-think-it-is-20241118/>



So how can one get better at **stats**?

Whenever you **spot a disconnect** between what your gut is telling you and what is supposed to be rational, it's an important **opportunity to understand something new**. And then you can start this game of back-and-forth:

Can you articulate your gut instinct and **formalize it as a statistical model**?
If there's still a disconnect, can you **simulate and visualize your thinking**?

As you **play that game**, your imagination will gradually reconfigure. And in the end, if you're persistent, your **intuitions** and your reason will align...this is **statistical thinking**

<https://www.quantamagazine.org/mathematical-thinking-isnt-what-you-think-it-is-20241118/>

So what is a **statistical** model?

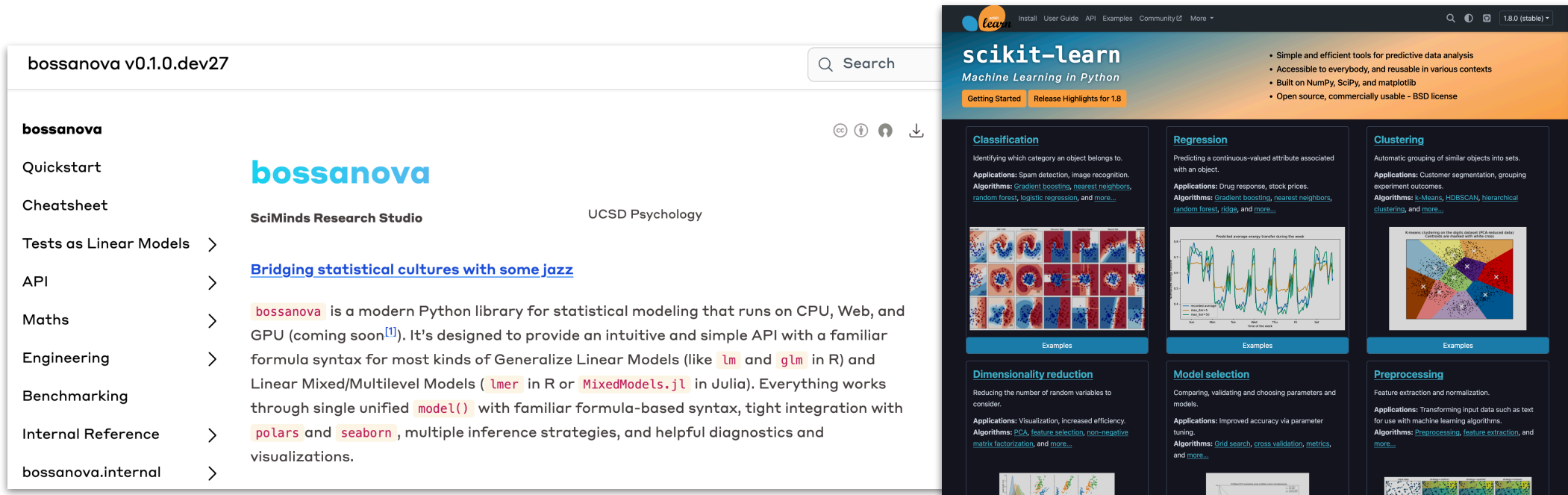
A **theory** of how **observed data** were **generated**

$$\text{Data} = \text{Model} + \text{Error}$$

*A model is a **logical story** expressed in mathematics*

~ Andrew Gelman

Studying Library Documentation is a cheat-code!



The image displays two web pages side-by-side. The left page is the documentation for 'bossanova v0.1.0.dev27'. It features a sidebar with navigation links: Quickstart, Cheatsheet, Tests as Linear Models, API, Maths, Engineering, Benchmarking, Internal Reference, and bossanova.internal. The main content area shows the 'bossanova' logo, the text 'SciMinds Research Studio' and 'UCSD Psychology', and a link to 'Bridging statistical cultures with some jazz'. Below this, a paragraph describes bossanova as a modern Python library for statistical modeling that runs on CPU, Web, and GPU, designed to provide an intuitive and simple API with a familiar formula syntax for various models like Linear Mixed/Multilevel Models, and Linear Mixed/Multilevel Models. It mentions integration with 'polars' and 'seaborn' for inference strategies and diagnostics and visualizations.

The right page is the 'scikit-learn' documentation, titled 'Machine Learning in Python'. It includes a navigation bar with links like 'Install', 'User Guide', 'API', 'Examples', 'Community', and 'More'. The main content area is divided into several sections: 'Classification' (Identifying which category an object belongs to, with applications like spam detection and image recognition), 'Regression' (Predicting a continuous-valued attribute associated with an object, with applications like drug response and stock prices), 'Clustering' (Automatic grouping of similar objects into sets, with applications like customer segmentation), 'Dimensionality reduction' (Reducing the number of random variables to consider, with applications like visualization and increased efficiency), 'Model selection' (Comparing, validating and choosing parameters and models, with applications like improved accuracy via parameter tuning), and 'Preprocessing' (Feature extraction and normalization, with applications like transforming input data such as text for use with machine learning algorithms). Each section includes a brief description, applications, algorithms, and a small example image.

Keep stat-intuitions.com bookmarked through your PhD journey!

We'll keep it updated as a Department Resource

2) Adopt a “hacker” mindset

Statistics for Hackers

Hacker (*n.*)

1. ~~A person who is trying to steal your grandma's bank password.~~
2. A person whose natural approach to problem-solving involves writing code.

Statistics is Hard.

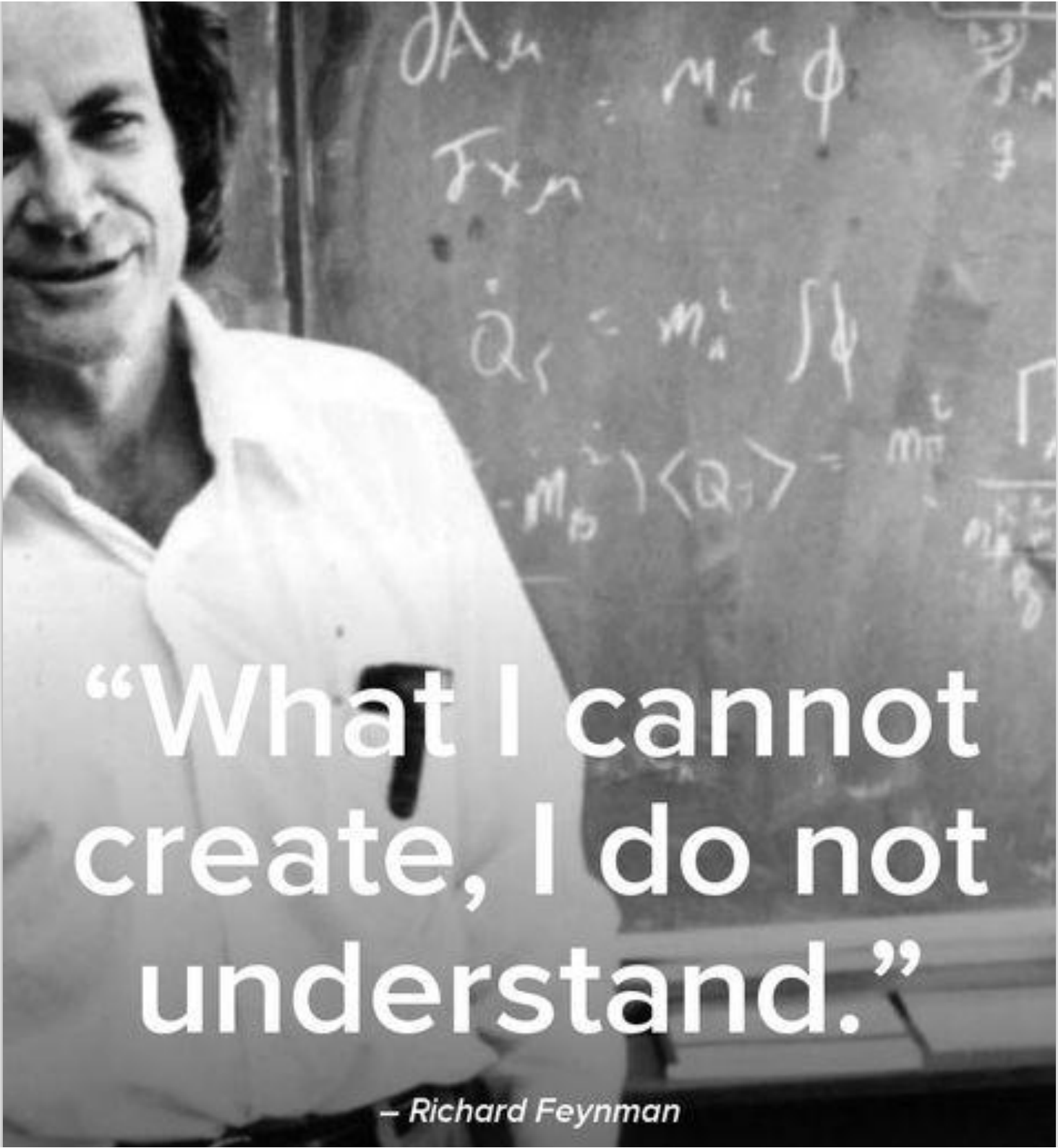
**Using programming skills,
it can be easy.**

Statistics is fundamentally about

Asking the Right Question.

Which means

**If you can write a for-loop,
you can do statistics**

A black and white photograph of Richard Feynman. He is smiling and looking towards the camera. He is wearing a light-colored, possibly white, button-down shirt. Behind him is a chalkboard filled with handwritten mathematical equations. The equations include $\partial A_\mu = m_\pi^2 \phi$, $\mathcal{F} \times n$, $\dot{Q}_\zeta = m_\pi^2 \int \psi$, and $(-m_\pi^2) \langle Q \rangle = m_\pi^2$.

**“What I cannot
create, I do not
understand.”**

– Richard Feynman

Programming as theory-building

A program (Python/R) is simply a *hypothesis made executable*

PETER NAUR, PROGRAMMING AS THEORY BUILDING

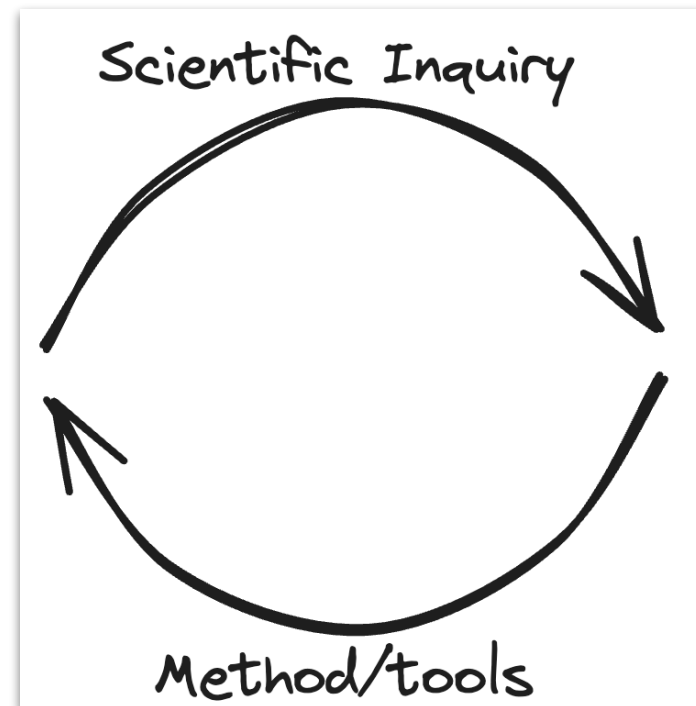
Peter Naur, widely known as one of the authors of the programming language syntax notation "Backus-Naur Form" (BNF), wrote "Programming as Theory Building" in 1985. It was reprinted in his collection of works, *Computing: A Human Activity* (Naur 1992).

This article is, to my mind, the most accurate account of what goes on in designing and coding a program. I refer to it regularly when discussing how much documentation to create, how to pass along tacit knowledge, and the value of the XP's metaphor-setting exercise. It also provides a way to examine a methodology's economic structure.

"PROGRAMMING AS THEORY BUILDING"

Introduction

The present discussion is a contribution to the understanding of what programming is. It suggests that programming properly should be regarded as an activity by which the programmers form or achieve a certain kind of insight, a theory, of the matters at hand. This suggestion is in contrast to what appears to be a more common notion, that programming should be regarded as a production of a program and certain other texts.



How to be a Wizard Programmer

who can do anything (takes a very long time)

① ASK QUESTIONS. As long as there are people around you who know things you don't, ask them how to do things. Dumb questions. Scary-to-ask questions. Your questions will get less dumb fast.

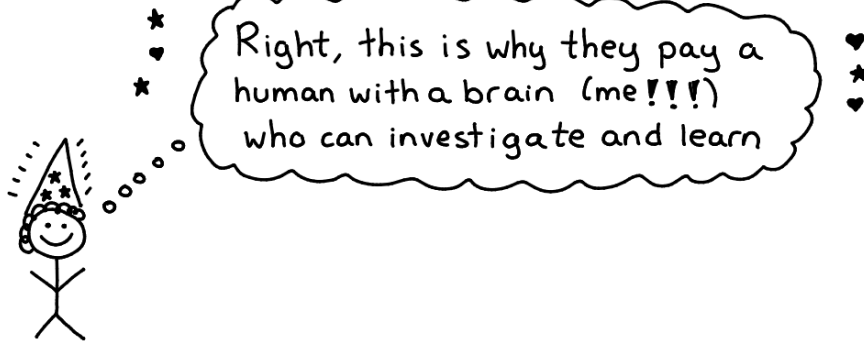
② Run into a problem your coworkers don't know how to solve either.

③ DECIDE YOU WILL FIGURE OUT HOW TO SOLVE THE PROBLEM ANYWAY
(this is very hard but sometimes it works ☺)

The more programming I do, the more issues I run into where:

- I don't know
- my colleagues don't know
- Google doesn't know
- we gotta figure it out anyway

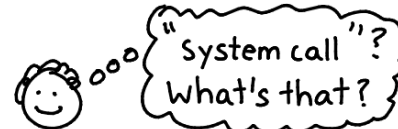
When this happens, I think:



This zine is about what the skill of "figure it out anyway" looks like.

ways to build expertise

learn fundamental concepts



- ① figure out which ideas are the most important
- ② Learn them!

do experiments!



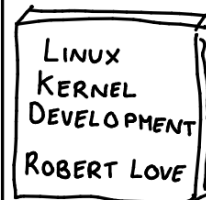
what system calls is THIS program using?

what's in /proc?

What happens if I run out of memory on purpose?

how long does it take to read 5GB from disk?

read books



NETWORKING FOR SYSTEM ADMINISTRATORS
MICHAEL LUCAS



even just reading a few chapters of a good book can help!

do hard projects



ooo ooh I'd need to learn a lot more to do that project

I'll work on that!

When you don't understand something, dig in



that's weird...

* figure it out *

I learned something new!!!

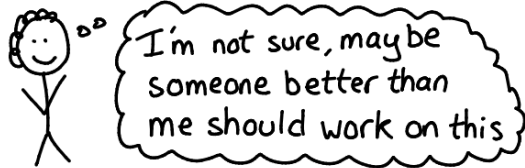
don't forget: it takes a long time



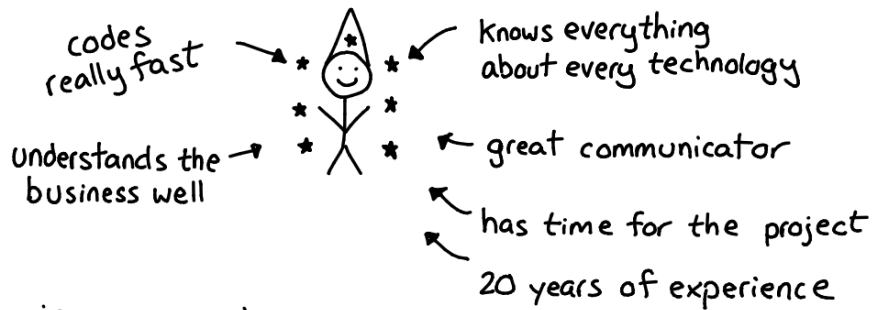
after 3 years I know a lot... but there's a lot more still!!!

take on hard projects

To wrap up, let's talk about one last wizard skill: confidence
When there's a hard project, sometimes I think:



and I imagine this ★ magical ★ human:



in programming:

- we're changing the tech we use all the time
- every project is different and it's rarely obvious how to do it
- there aren't many experts and they certainly don't have time to do everything.

So instead, I take myself:

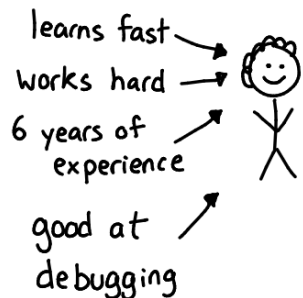
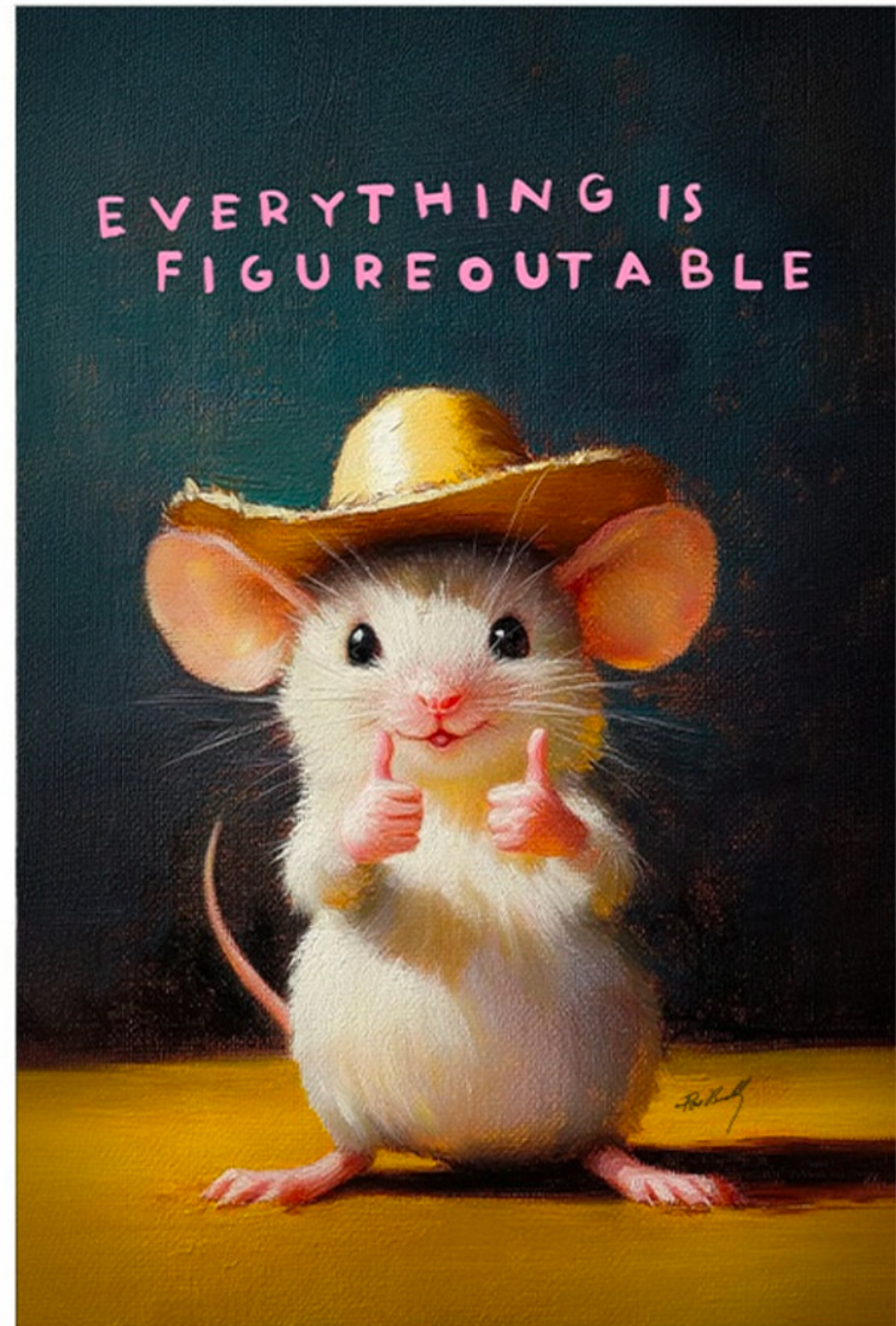
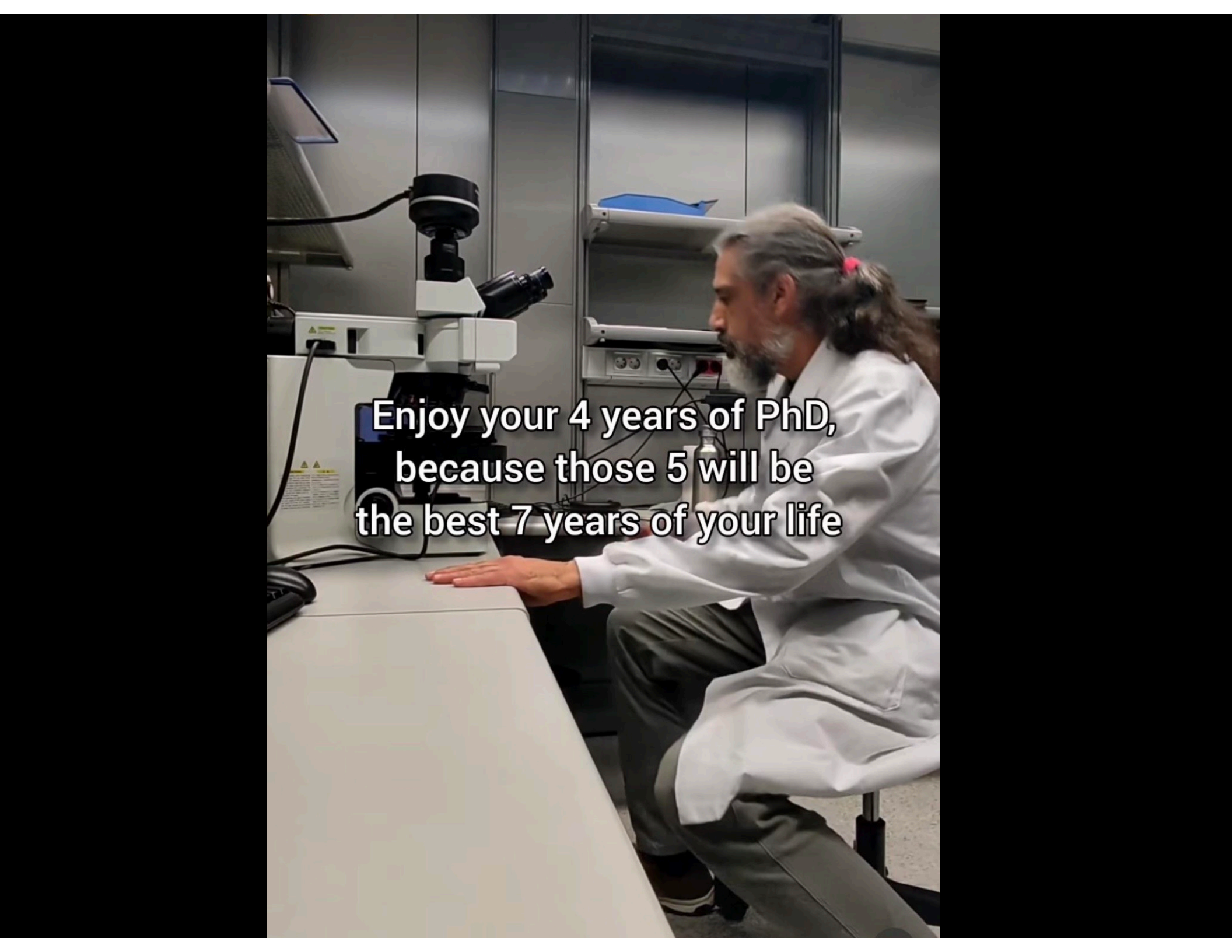


figure "someone's gotta do this",
write down a plan, and get
started! A lot of the time it
turns out well, I learn something,
and feel a little more like a
Wizard ♥



3) But mostly

A man with a grey beard and long hair tied back with a red hair tie is sitting at a white lab bench. He is wearing a white lab coat over grey pants. His right hand is resting on the bench. To his left is a large, white and black microscope. The background shows a laboratory environment with metal shelving and equipment. The text is overlaid in the center of the image.

**Enjoy your 4 years of PhD,
because those 5 will be
the best 7 years of your life**

Thanks to your **awesome** TA!



Jane Yang

And Thank YOU!

- We all had a lot of unexpected hiccups through-out the quarter so we **super appreciate** your patience and flexibility!
- Please take 10 min to complete the Course Feedback Survey: academicaffairs.ucsd.edu/Modules/Evals/
 - Your feedback is incredibly important to us and directly helps improve new versions!
- Then feel free to work on your final project!